



Vijay Academy Presents



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Data Science and Big Data Analytics (DSBDA)

TE Computer Engineering (2019 Pattern)

One Shot Lecture

Unit 5. Big Data Analytics and Model Evaluation

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions



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Unit 5: Big Data Analytics and Model Evaluation

Syllabus Topics:

1. K-Means Clustering
2. Hierarchical Clustering
3. Time Series Analysis
4. Introduction to Text Analysis
5. Text Preprocessing
6. Bag of Words
7. TF-IDF
8. Social Network Analysis
9. Business Analysis
10. Model Evaluation and Selection
11. Metrics for Evaluating Classifier Performance
12. Confusion Matrix
13. AUC-ROC Curves
14. Holdout Method and Random Subsampling
15. K-Fold Cross Validation
16. Parameter Tuning and Optimization
17. Elbow Plot

1: K-Means Clustering

What is K-Means Clustering?

K-Means Clustering is an **unsupervised machine learning algorithm** that divides data into **K number of clusters** based on similarity.

Simply put — *grouping similar data points together into K groups automatically.*

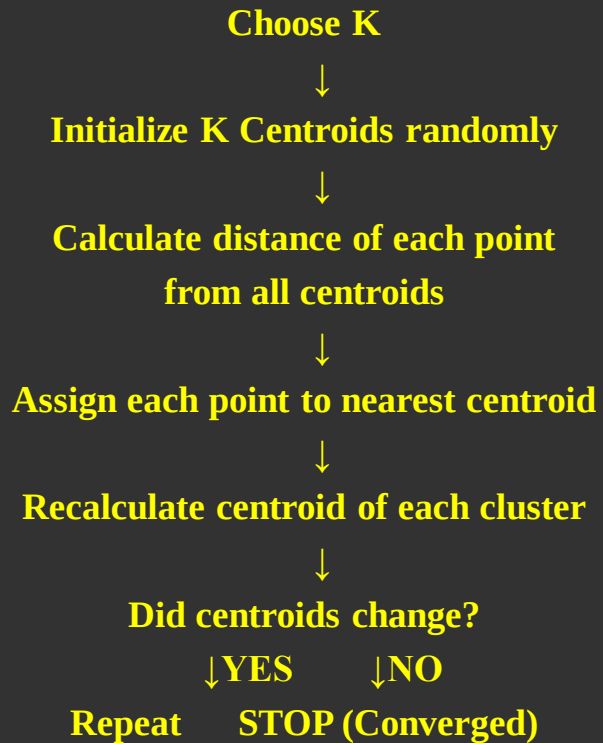
Key Terms:

Term	Meaning
K	Number of clusters to form
Centroid	Center point of each cluster
Cluster	Group of similar data points
Euclidean Distance	Distance between two points

Euclidean Distance Formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1: K-Means Clustering



1: K-Means Clustering

Advantages of K-Means:

Simple and easy to understand.

Fast for large datasets.

Works well when clusters are **spherical**.

Easy to **implement**.

Real Life Applications:

Customer segmentation — group customers by behavior

Image compression — reduce colors in image

Document clustering — group similar documents

Anomaly detection — find unusual patterns

2: Hierarchical Clustering

What is Hierarchical Clustering?

Hierarchical Clustering is a **clustering algorithm** that builds a **hierarchy of clusters** by either merging small clusters into bigger ones or splitting big clusters into smaller ones.

Simply put — *grouping data step by step into a tree like structure of clusters.*

Types of Hierarchical Clustering:

1. Agglomerative (Bottom-Up)

Starts with **each point as its own cluster**.

Merges closest clusters step by step.

Continues until **all points are in one cluster**.

Most commonly used approach.

2. Divisive (Top-Down)

Starts with **all points in one cluster**.

Splits clusters step by step.

Continues until **each point is its own cluster**.

Less commonly used.

Key Term - Dendrogram:

A Dendrogram is a **tree shaped diagram** that shows how clusters are merged or split at each step.
Simply put — *a visual representation of hierarchical clustering process.*

Real Life Applications:

Biology — grouping organisms by genetic similarity

Document clustering — grouping similar documents

Customer segmentation — grouping customers by behavior

Image segmentation — grouping similar pixels

3: Time Series Analysis

What is Time Series Analysis?

Time Series Analysis is a method of **analyzing data points collected over time** to find patterns, trends and make future predictions.

Simply put — *studying data that changes over time to understand patterns and predict future values.*

What is Time Series Data?

Data that is **recorded at regular time intervals** is called Time Series Data.

Examples:

Daily stock prices

Monthly sales figures

Hourly temperature readings

Weekly website traffic

Yearly population growth

Steps of Time Series Analysis:

Step 1: Collect Time Series Data

Gather data at **regular intervals**

Example: Daily stock prices for 1 year

Step 2: Visualize Data

Plot data to see **trends and patterns**

Use line charts

Step 3: Check Stationarity

Stationary data has constant mean and variance over time

Non stationary data needs to be **transformed**

Step 4: Decompose Time Series

Separate **trend, seasonality and noise**

Step 5: Build Forecasting Model

Apply **ARIMA or other models**

Train on historical data

Step 6: Make Predictions

Forecast **future values**

Evaluate model accuracy

4: Introduction to Text Analysis

What is Text Analysis?

Text Analysis is the process of **extracting useful information and patterns from raw text data** using computational methods.

Simply put — teaching computers to read, understand and find meaning in text.

Examples of text data:

Tweets, Facebook posts

Customer reviews

Emails, news articles

Chat messages



Steps in Text Analysis :

Step 1: Text Collection

Gather raw text data from sources like social media, websites, databases.

Step 2: Text Preprocessing

Clean and prepare the text for analysis. (We'll cover this in detail in next topic)

Step 3: Feature Extraction

Convert text into numbers that computers can understand. Example: Bag of Words, TF-IDF

Step 4: Model Building

Apply machine learning algorithms on extracted features.

Step 5: Result Interpretation

Understand and use the results.

Applications of Text Analysis

Application	Example
Sentiment Analysis	Is this review positive or negative?
Spam Detection	Is this email spam or not?
Language Translation	English to Marathi
Chatbots	Customer support bots
Search Engines	Google search results
News Classification	Sports, Politics, Tech

5: Text Preprocessing

What is Text Preprocessing?

Text Preprocessing is the process of **cleaning and preparing raw text data** before applying any analysis or machine learning algorithm.

Simply put — raw text is messy, preprocessing makes it clean and ready to use.

Example:

Raw text: "The QUICK brown fox!! Jumped over... THE lazy dog!!!"

After preprocessing: "quick brown fox jump lazy dog"



Steps of Text Preprocessing

Step 1: Lowercasing

Convert all text to lowercase so "Apple" and "apple" are treated as same word.

Example:

Before: "Data Science is GREAT"

After: "data science is great"

Step 2: Removing Punctuation & Special Characters

Remove symbols like ! ? , . # @ etc.

Example:

Before: "Hello!! How are you???"

After: "Hello How are you"

Step 3: Removing Stop Words

Stop words are common words that carry no useful meaning like "is", "the", "a", "an", "in", "on".

Example:

Before: "the cat is sitting on the mat"

After: "cat sitting mat"

Step 4: Tokenization

Splitting text into individual words or sentences called **tokens**.

Example:

Before: "Data science is fun"

After: ["Data", "science", "is", "fun"]

Steps of Text Preprocessing



Step 5: Stemming

Reducing a word to its **root form** by removing suffixes. The result may not be a real word.

Example:

Original	After Stemming
running	run
jumped	jump
playing	play
studies	studi
history	histori

Step 6: Lemmatization

Similar to stemming but always gives a **real dictionary word**.

Example:

Original	After Lemmatization
running	run
better	good
studies	study
history	history

Step 7: POS Tagging (Part of Speech Tagging)

Identifying the grammatical role of each word — noun, verb, adjective etc.

Example:

Sentence: "John runs fast"

John → Noun

runs → Verb

fast → Adverb



6: Bag of Words

What is Bag of Words?

Bag of Words (BoW) is a technique to **convert text into numbers** so that machine learning algorithms can understand it.

Simply put — count how many times each word appears in a document and represent it as a table of numbers.

The word "bag" means we **don't care about word order** — just the count of each word matters.

Example :

"I love cricket. I love football." $\rightarrow$ bag = {I:2, love:2, cricket:1, football:1}

Steps to Build Bag of Words

Step 1: Collect all documents (sentences)

Step 2: Build a vocabulary — list of all unique words across all documents

Step 3: For each document, count how many times each vocabulary word appears

Step 4: Represent each document as a vector of counts

6: Bag of Words

Complete Example

Documents:

D1: "I love cricket"

D2: "I love football"

D3: "cricket is great"

Step 1 — Build Vocabulary (all unique words): $\rightarrow \{I, \text{love}, \text{cricket}, \text{football}, \text{is}, \text{great}\}$

Step 2 — Create BoW Matrix:

Document	I	love	cricket	football	is	great
D1	1	1	1	0	0	0
D2	1	1	0	1	0	0
D3	0	0	1	0	1	1

Each row = one document represented as numbers.

7: TF-IDF

What is TF-IDF?

TF-IDF stands for **Term Frequency - Inverse Document Frequency**.

It is a technique to find **how important a word is** in a document compared to all other documents.

Simply put — common words like "the", "is" get low score. Rare but important words get high score.

Part 1: TF (Term Frequency)

How many times a word appears in **one document**.

$TF = \text{Number of times word appears in document} / \text{Total words in document}$

Part 2: IDF (Inverse Document Frequency)

How rare a word is **across all documents**.

$IDF = \log\left(\frac{\text{Total number of documents}}{\text{Number of documents containing the word}}\right)$

Final Formula:

$TF-IDF = TF \times IDF$

Complete Numerical Example



Documents:

D1: "cricket is a great sport"

D2: "football is a great sport"

D3: "cricket and football are popular"

calculate TF-IDF for word "is" in D1:

Step 1 — TF:

"is" appears 1 time in D1

Total words = 5

$$TF = 1/5 = 0.2$$

Step 2 — IDF:

Documents containing "is" = D1 and D2 = 2

$$IDF = \log(3/2) = 0.176$$

Step 3 — TF-IDF:

$$TF-IDF = 0.2 \times 0.176 = 0.035$$

Now calculate TF-IDF for word "popular" in D3:

Step 1 — TF:

"popular" appears 1 time in D3

Total words in D3 = 5

$$TF = 1/5 = 0.2$$

Step 2 — IDF:

Documents containing "popular" = only D3 = 1

$$IDF = \log(3/1) = \log(3) = 0.477$$

Step 3 — TF-IDF:

$$TF-IDF = 0.2 \times 0.477 = 0.095$$

"popular" gets higher score than "is" because it is **rarer** and more important!

8: Social Network Analysis

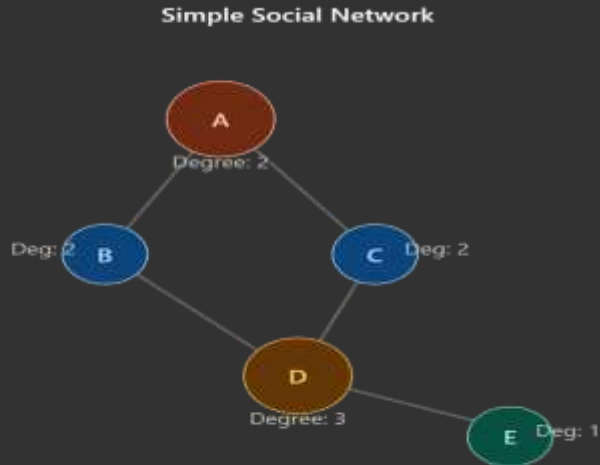
What is Social Network Analysis?

Social Network Analysis (SNA) is the study of **relationships and connections between people or entities** in a network.

Simply put — analyzing who is connected to whom and how information flows between them.

Example:

On Instagram — you follow people, people follow you, you tag friends. All these connections form a **social network**.



Key Concepts

- High centrality node
- Regular node
- Hub node (D, deg=3)
- Leaf node (E, deg=1)
- Edge (connection)

Network Stats

Total nodes: 5
 Total edges: 5
 Most connected: D

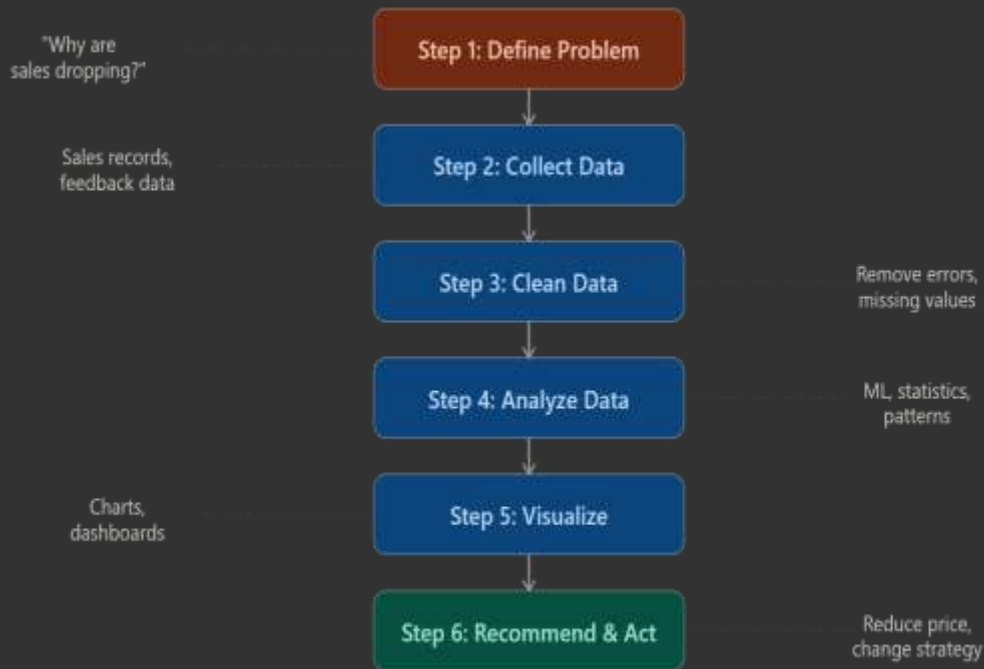
Real Life Applications

Application	Example
Finding influencers	Who has most followers on Twitter
Epidemic spreading	How disease spreads person to person
Recommendation systems	Suggest friends on Facebook
Crime detection	Finding connections between criminals
Marketing	Identify key people to spread a message
Fake news detection	How mis-information spreads

9: Business Analysis

What is Business Analysis?

Business Analysis is the process of **using data to understand business performance, identify problems and make better business decisions.**





10: Model Evaluation and Selection

What is Model Evaluation?

Model Evaluation is the process of **measuring how well a machine learning model performs** on data.

Simply put — after building a model, we need to check: "Is this model good enough? Is it giving correct predictions?"

Why is Model Evaluation Important?

A model may perform well on training data but poorly on new data

We need to select the **best model** from multiple options

Helps detect **overfitting** and **underfitting**

Ensures model is Dependable, trustworthy before deploying

Model Selection

After evaluating multiple models, we **select the best one** based on:

Highest accuracy

Lowest error

Best balance between overfitting and underfitting

Speed and simplicity

11: Metrics for Evaluating Classifier Performance

What are Evaluation Metrics?

Evaluation Metrics are **numbers that tell us how well a classification model is performing.**

Simply put — after building a model, we use these metrics to give it a "score" — like marks in an exam.

First Understand: The 4 Outcomes

When a model predicts Yes or No, there are 4 possible outcomes:

	Predicted YES	Predicted NO
Actual YES	TP (True Positive)	FN (False Negative)
Actual NO	FP (False Positive)	TN (True Negative)



All Metrics with Formulas

1. Accuracy

How many predictions were correct overall?

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

2. Precision

Out of all predicted YES, how many were actually YES?

$$\text{Precision} = \frac{TP}{TP+FP}$$

3. Recall (Sensitivity)

Out of all actual YES, how many did we correctly predict?

$$\text{Recall} = \frac{TP}{TP+FN}$$

4. Error Rate

How many predictions were wrong?

$$\text{Error Rate} = 1 - \text{Accuracy}$$

5. F1 Score

Balance between Precision and Recall

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

11: Metrics for Evaluating Classifier Performance

Confusion Matrix — 4 Outcomes

	Predicted YES	Predicted NO
Actual YES	TP True Positive	FN False Negative
Actual NO	FP False Positive	TN True Negative

Metrics Summary

Accuracy $(TP+TN) / \text{Total}$	Precision $TP / (TP+FP)$	Recall $TP / (TP+FN)$	Error Rate $(FP+FN) / \text{Total}$
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When to use which metric?

Accuracy → balanced data | Precision → minimize false alarms | Recall → minimize missed cases

12: Confusion Matrix

What is a Confusion Matrix?

A Confusion Matrix is a **table that shows the performance of a classification model** by comparing actual values vs predicted values.

Simply put — it shows us exactly where the model got confused (made mistakes).

Structure of Confusion Matrix :

For a binary classification (Yes/No, Spam/Not Spam):

	Predicted YES	Predicted NO
Actual YES	TP (True Positive)	FN (False Negative)
Actual NO	FP (False Positive)	TN (True Negative)

13: AUC-ROC Curve

What is AUC-ROC?

ROC = Receiver Operating Characteristic Curve

AUC = Area Under the Curve

Simply put — ROC is a **graph** that shows how well a classification model performs.

AUC is the **single number score** that summarizes that graph.

Two Key Terms

1. TPR (True Positive Rate) = Recall = Sensitivity

How many actual positives did we correctly catch?

$$\text{TPR} = \text{TP} / \text{TP} + \text{FN}$$

2. FPR (False Positive Rate)

How many actual negatives did we wrongly classify as positive?

$$\text{FPR} = \text{FP} / \text{FP} + \text{TN}$$

13: AUC-ROC Curve

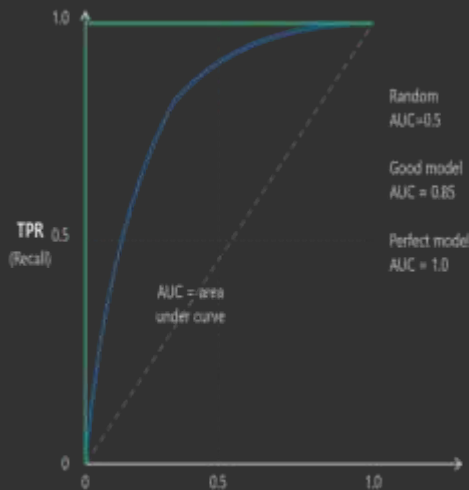
How ROC Curve is Plotted

X axis = FPR (False Positive Rate) $\rightarrow$ 0 to 1

Y axis = TPR (True Positive Rate) $\rightarrow$ 0 to 1

Each point on curve = one threshold value

As threshold changes, TPR and FPR change



AUC Score Interpretation

AUC=1.0 Perfect | AUC=0.9 Excellent | AUC=0.7-0.8 Good | AUC=0.5 Random | AUC<0.5 Bad

Real Life Applications

Medical diagnosis — is this model good at detecting cancer?

Spam detection — how well does filter separate spam from real emails?

Credit scoring — how well does model predict loan defaults?

Fraud detection — how well does model catch fraudulent transactions?

14: Holdout Method and Random Subsampling

What is the Holdout Method?

The Holdout Method is a model evaluation technique where the dataset is **split into two parts** — one for training and one for testing.

Simply put — you hide some data from the model during training, then use that hidden data to test how well the model learned.

Simple Analogy

Think of it like studying for an exam:

You study from your textbook (training data) → You give the exam with new questions (test data) → Your marks = model accuracy

Steps of Holdout Method

Step 1: Take the full dataset

Step 2: Split into Training set and Test set

Common split: **70% training, 30% testing**

Other splits: 80-20, 60-40

Step 3: Train the model on training set only

Step 4: Test the model on test set

Step 5: Calculate accuracy and other metrics

14: Holdout Method and Random Subsampling

What is Random Subsampling?

Random Subsampling is an **improved version of the Holdout Method** where the train-test split is done **multiple times randomly** and the final accuracy is the average of all runs.

Simply put — instead of splitting data once, split it randomly many times and average the results.

Steps of Random Subsampling

Step 1: Take full dataset

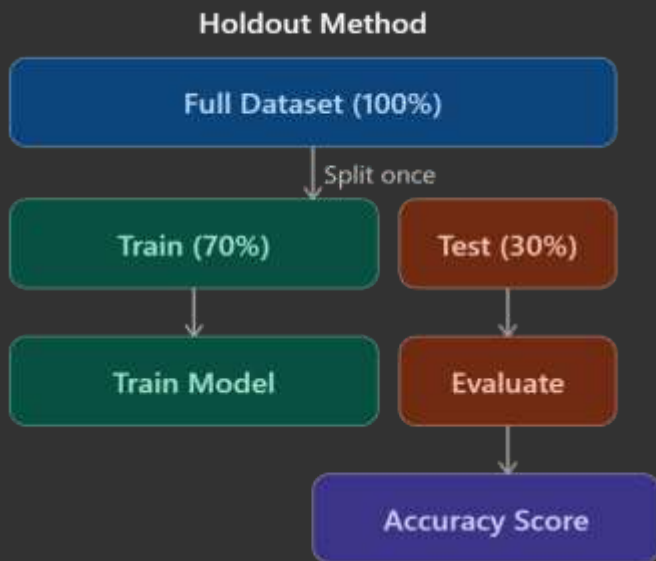
Step 2: Randomly split into train and test (e.g. 70-30)

Step 3: Train model and record accuracy

Step 4: Repeat Steps 2 and 3 N times (e.g. 10 times)

Step 5: Final accuracy = Average of all N accuracies

14: Holdout Method and Random Subsampling



Holdout — Pros and Cons

- + Simple and fast
- + Easy to understand
- Result depends on how you split



Random Subsampling — Pros and Cons

- + More reliable than holdout
- + Reduces bias from one split
- Same data may appear multiple times

15: K-Fold Cross Validation

What is K-Fold Cross Validation?

K-Fold Cross Validation is a model evaluation technique where the dataset is **divided into K equal parts (folds)** and the model is trained and tested K times — each time using a different fold as the test set.

Simply put — instead of one fixed train-test split, we rotate the test set K times so every part of data gets a chance to be tested.

Steps of K-Fold Cross Validation

Step 1: Choose value of K (commonly $K=5$ or $K=10$)

Step 2: Divide dataset into K equal folds

Step 3: In Round 1 → Use Fold 1 as test, remaining $K-1$ folds as training

Step 4: Train model, record accuracy

Step 5: In Round 2 → Use Fold 2 as test, remaining as training

Step 6: Repeat until all K folds have been used as test set once

Step 7: Final accuracy = Average of all K accuracies

15: K-Fold Cross Validation

Example with K=5

K-Fold Cross Validation (K=5)						
Round	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Acc
Round 1	TEST	Train	Train	Train	Train	88%
Round 2	Train	TEST	Train	Train	Train	85%
Round 3	Train	Train	TEST	Train	Train	90%
Round 4	Train	Train	Train	TEST	Train	87%
Round 5	Train	Train	Train	Train	TEST	86%

Final Accuracy = $(88+85+90+87+86)/5 = 87.2\%$
Average of all K rounds

Train

TEST

Each fold gets one chance to be the test set

16: Parameter Tuning and Optimization

What is Parameter Tuning?

Parameter Tuning is the process of **finding the best settings (parameters) for a machine learning model** to get the highest possible performance.

Simply put — every ML model has some settings (like knobs on a radio). Parameter tuning means adjusting those knobs to get the best output.

Two Types of Parameters

1. Model Parameters

Learned **automatically** during training

Example: weights in a neural network, coefficients in linear regression

You don't set these manually

2. Hyperparameters

Set **manually before** training begins

Example: K in K-Means, depth of decision tree, learning rate

These are what we **tune**

16: Parameter Tuning and Optimization

What is Optimization?

Optimization is the process of **minimizing error or maximizing accuracy** of a model during training. Simply put — finding the best possible model by reducing mistakes.

Most common optimization technique:

Gradient Descent :

1. Starts with random weights
2. Calculates error
3. Adjusts weights to reduce error
4. Repeats until error is minimum

17: Elbow Plot



What is an Elbow Plot?

An Elbow Plot is a graph used to **find the optimal value of K** (number of clusters) in K-Means Clustering. Simply put — it helps answer the question: "**How many clusters should I make?**"

The Problem it Solves

In K-Means, we need to decide K before starting. But how do we know the best K?

K=1 → all points in one cluster → too simple

K=10 → too many clusters → overfitting

Elbow Plot helps find the sweet spot!

Key Term: WCSS

WCSS = Within Cluster Sum of Squares

WCSS measures how **compact** the clusters are — how close all points are to their cluster centroid.

$$WCSS = \sum_{i=1}^K \sum_{x \in C_i} (x - \mu_i)^2$$

17: Elbow Plot



Steps to Create Elbow Plot

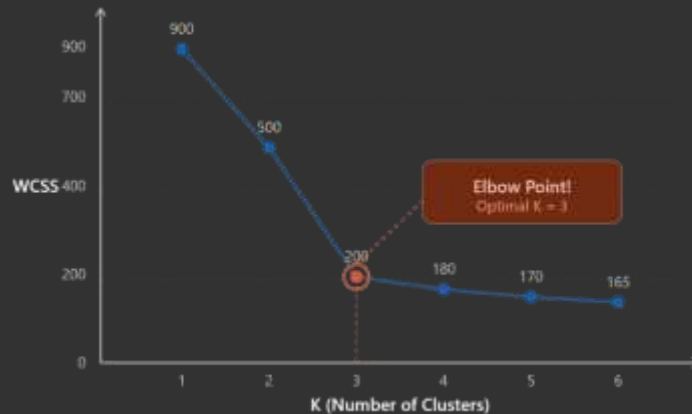
Step 1: Run K-Means for $K=1, 2, 3, 4, 5...$ (different values of K)

Step 2: For each K , calculate WCSS

Step 3: Plot K on X-axis and WCSS on Y-axis

Step 4: Look for the "elbow point" where WCSS stops decreasing rapidly

Step 5: That K value = optimal number of clusters.



After $K=3$, WCSS barely decreases — No benefit in adding more clusters → $K=3$ is optimal



Thank You!



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